Lecture 3: Conditional Statements & Loops

DSA Series | Shradha Khapra (Apna College) | C++ | Short Notes

1. Conditional Statements (if - else)

- Program mein **decision** lena ho (ye karo ya wo karo) to conditional statements use karte hain. Flowchart mein ye **diamond** hota hai.
- **if:** condition **true** ho to hi block chalta hai. **else:** condition **false** ho to chalta hai.
- Condition mein relational (**< > == !=**) aur logical (**&& || !**) operators use hote hain. Condition ka result **true / false**.
- Block **{ }** ke andar likhte hain. Sirf ek statement ho to braces optional hain, par lagana safe hai.
- **Galti dhyan se:** `if (a = 5)` assign karta hai, compare nahi. Compare ke liye ==. Aur `if (...);` ke baad semicolon mat lagao.

```
int age = 20;
if (age >= 18) {
    cout << "Adult";
} else {
    cout << "Minor";
}
```

2. else if

- Do se zyada options ho to **else if ladder**. Conditions **upar se neeche** check hoti hain.
- **Pehli true condition** ka block chalta hai, baaki sab **skip**. Koi bhi true na ho to last ka **else** chalta hai.
- Condition ka order sahi rakho (e.g. marks >= 90 pehle, phir >= 75).

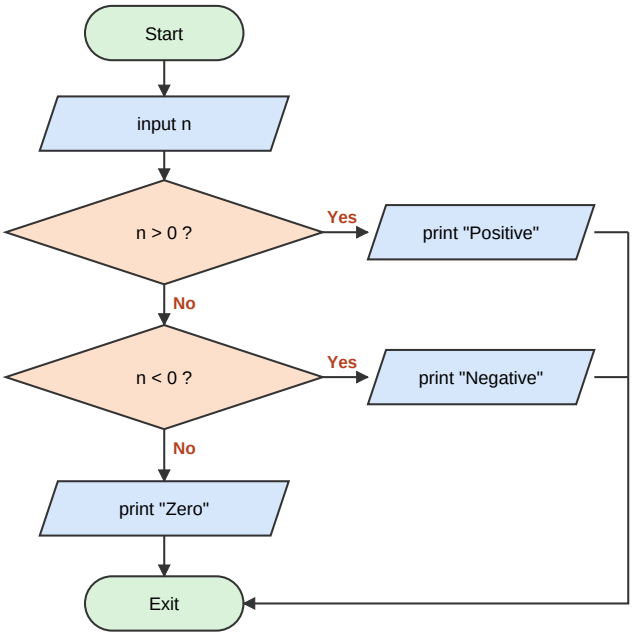
```
int marks = 82;
if (marks >= 90)      cout << "Grade A";
else if (marks >= 75) cout << "Grade B";
else if (marks >= 50) cout << "Grade C";
else                  cout << "Fail";   // Grade B print hoga
```

3. Practice Questions: if / else if (Flowchart + Pseudocode)

Q1: Positive, Negative ya Zero

- Do decisions: pehle n > 0, phir n < 0.
- Dono false ho to **Zero**.

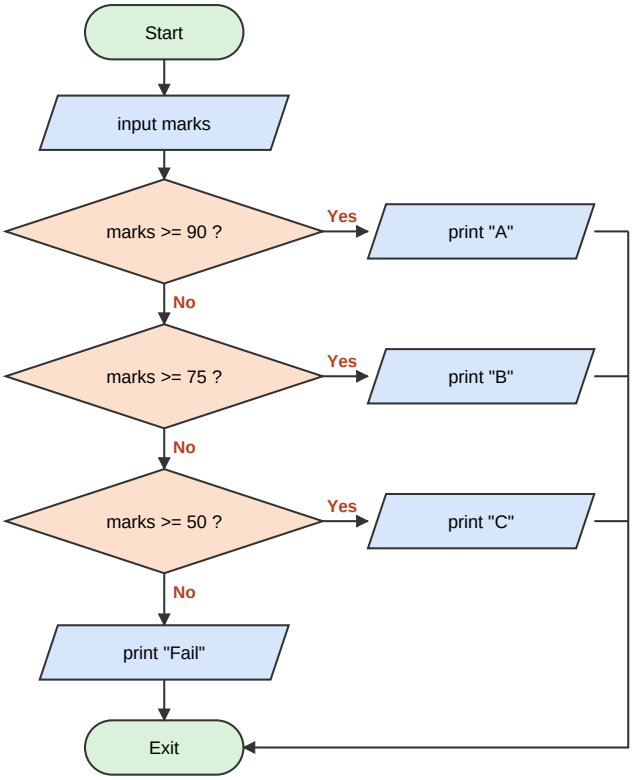
```
input n
if n > 0
    print "Positive"
else if n < 0
    print "Negative"
else
    print "Zero"
exit
```



Q2: Marks se Grade

- Ladder: >= 90 A, >= 75 B, >= 50 C, warna Fail.
- Example: marks = 82 -> 82 >= 90 false, 82 >= 75 true -> **B**.

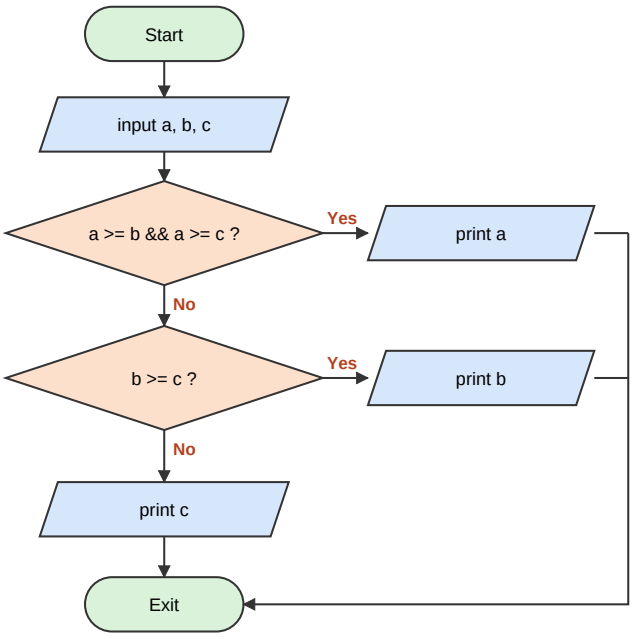
```
input marks
if marks >= 90
    print "A"
else if marks >= 75
    print "B"
else if marks >= 50
    print "C"
else
    print "Fail"
exit
```



Q3: Teen numbers mein largest

- **&&** se do conditions ek saath: a sabse bada tabhi jab a >= b **aur** a >= c.
- Nahi to b aur c mein compare.

```
input a, b, c
if a >= b && a >= c
    print a
else if b >= c
    print b
else
    print c
exit
```



4. Ternary Operator

- if-else ka **chhota rup**, ek hi line mein. Syntax: `condition ? value1 : value2`
- Condition **true** to value1, **false** to value2. Ye teen operands leta hai, isliye **ternary**.
- Sirf chhote, simple decisions ke liye. Lambe logic mein if-else hi sahi.

```
int a = 10, b = 5;
int max = (a > b) ? a : b;      // 10
cout << (a % 2 == 0 ? "Even" : "Odd");
```

5. while Loop

- Kaam **baar-baar** dohrana ho to **loop**. **while** mein condition **pehle** check hoti hai.
- Loop ke **3 hisse**: **initialization** (i = 1), **condition** (i <= n), **update** (i++).
- **Update bhool gaye** to condition kabhi false nahi hogi: **infinite loop**.
- Condition shuru mein hi false ho to loop **ek baar bhi nahi** chalta.
- Dry run karo: har iteration mein variables ki value likho.

```
int i = 1;              // initialization
while (i <= 5) {        // condition
    cout << i << " ";
    i++;                // update
}                       // output: 1 2 3 4 5
```

6. for Loop

- Initialization, condition aur update **ek hi line** mein: `for (init; condition; update)`. Count pata ho to for aasan padta hai.
- Chalne ka order: **init (sirf ek baar)** -> condition check -> body -> update -> condition check ...
- `for (int i = 1; ...)` mein **i sirf loop ke andar** milta hai (scope).
- **for aur while** ka logic aur flowchart same hai, bas likhne ka tareeka alag.

```
for (int i = 1; i <= 5; i++) {
    cout << i << " ";
}                       // output: 1 2 3 4 5
```

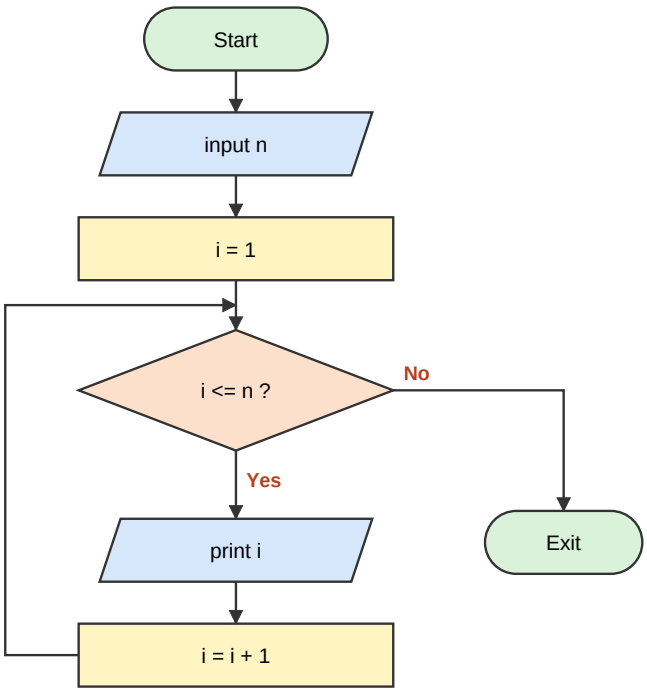
7. Practice Questions: Loops (Flowchart + Pseudocode)

Neeche ke flowcharts while aur for dono ke liye same hain.

Q4: 1 se n tak numbers print karo

- **i = 1** se shuru, jab tak **i <= n** print karo aur i badhao.

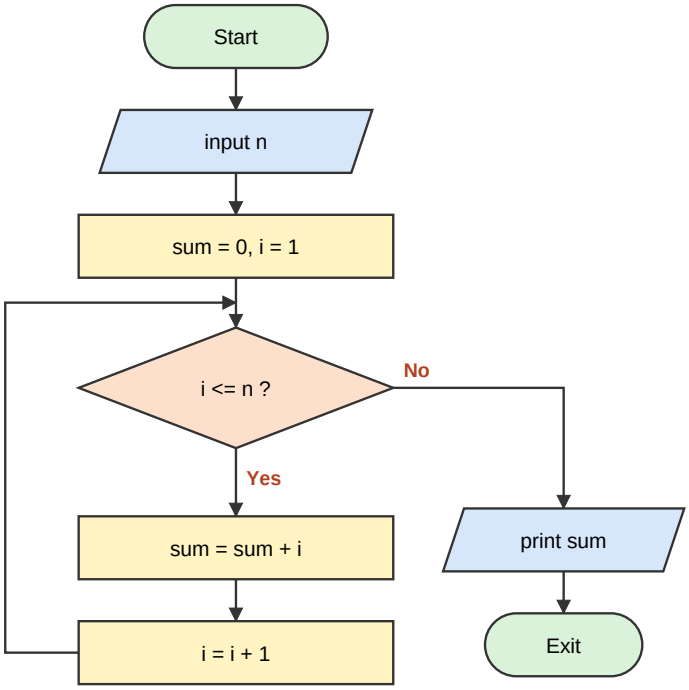
```
input n
i = 1
while i <= n
    print i
    i = i + 1
exit
```



Q5: 1 se n tak ka sum

- **sum = 0** se shuru (initialization zaroori).
- Dry run (n = 4): 1, 3, 6, **10**.

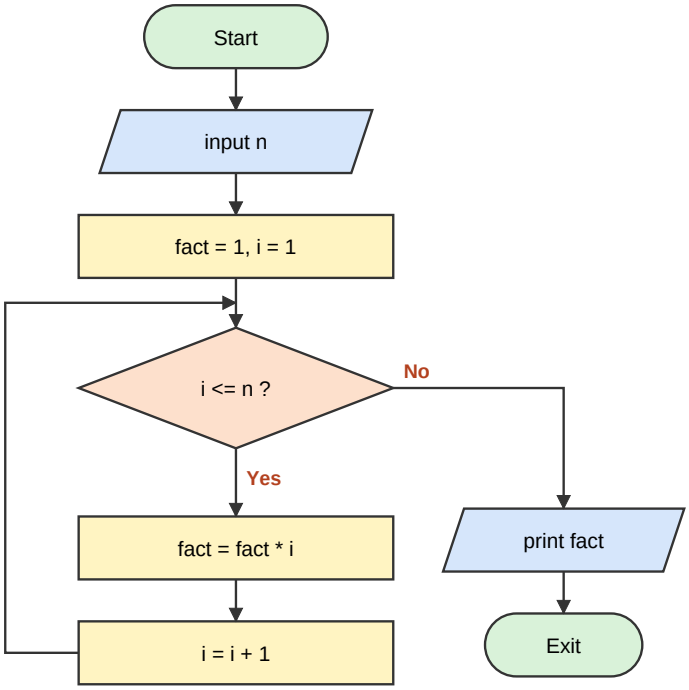
```
input n
sum = 0
i = 1
while i <= n
    sum = sum + i
    i = i + 1
print sum
exit
```



Q6: n ka factorial

- n! = 1 x 2 x ... x n.
- **fact = 1** se shuru (0 se nahi, warna sab 0 ho jayega).
- Dry run (n = 4): 1, 2, 6, **24**.

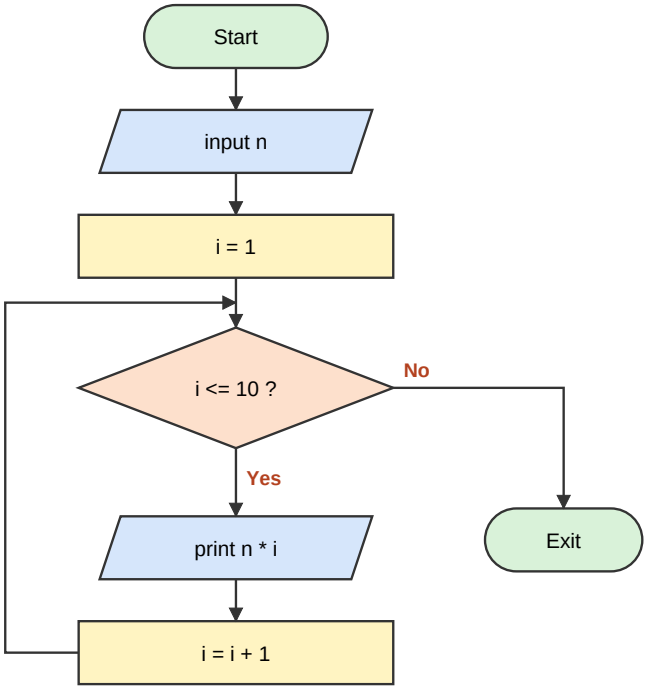
```
input n
fact = 1
for i = 1 to n
    fact = fact * i
print fact
exit
```



Q7: n ki multiplication table

- i = 1 se 10 tak **n x i** print karo.
- Example n = 5: 5, 10, 15, ... 50.

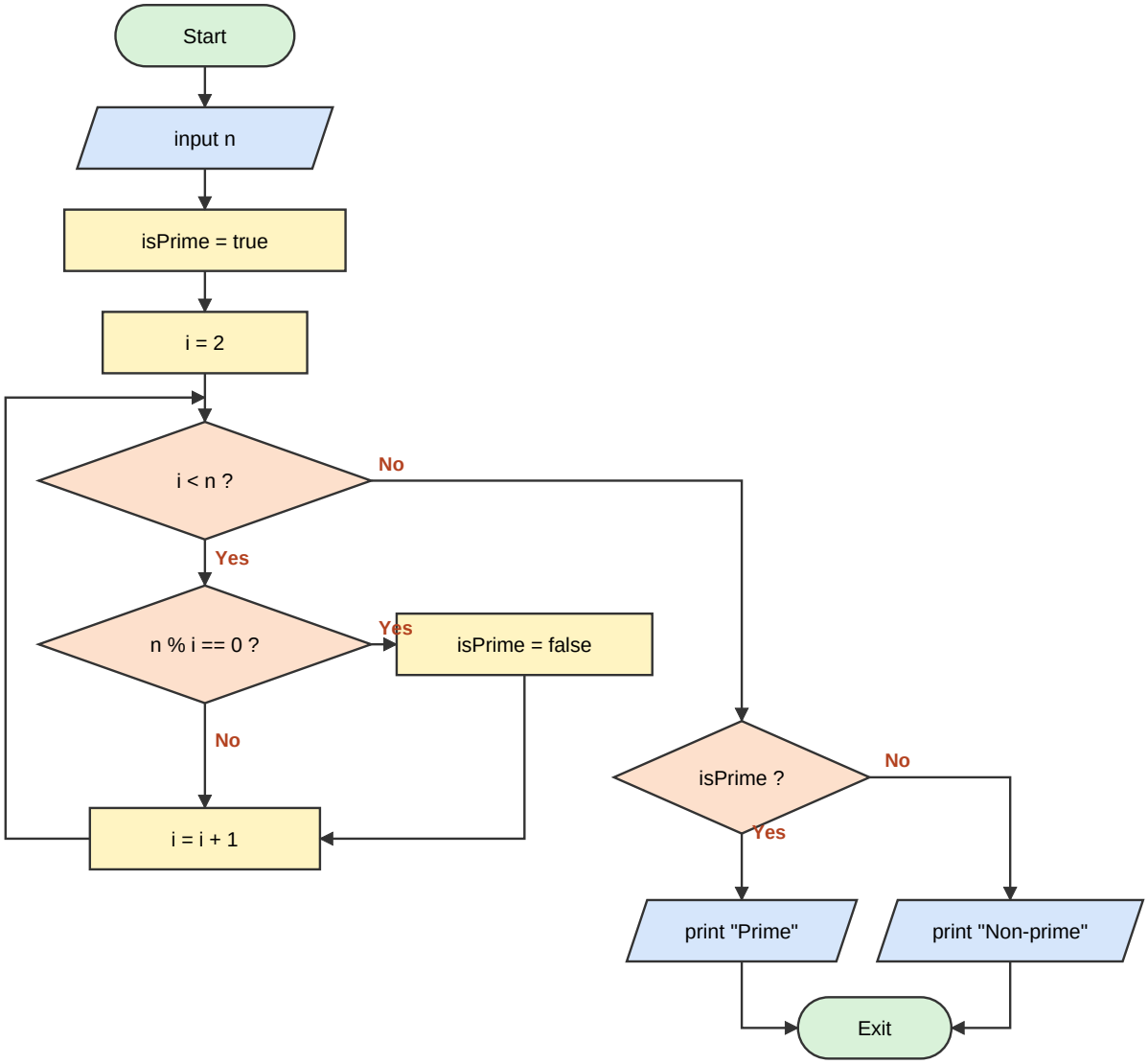
```
input n
for i = 1 to 10
    print n * i
exit
```



Q8: Prime number check (flag ke saath)

- **Flag** (isPrime) pehle **true** maan lo. i = 2 se n-1 tak koi bhi i n ko divide kare to flag **false**.
- Loop ke baad flag dekho: true -> Prime, false -> Non-prime. Loop ke andar **if** aata hai (loop + decision).
- **n <= 1** prime nahi hota, iska alag check lagao.

```
input n
isPrime = true
for i = 2 to n - 1
    if n % i == 0
        isPrime = false
if isPrime
    print "Prime"
else
    print "Non-prime"
exit
```



8. do-while Loop

- Condition **baad mein** check hoti hai, isliye body **kam se kam ek baar** zaroor chalti hai.
- Last mein **while (condition);** ke baad **semicolon** lagta hai.
- Jab pehle kaam karna ho phir dekhna ho dobara karna hai ya nahi (jaise menu, input validation).

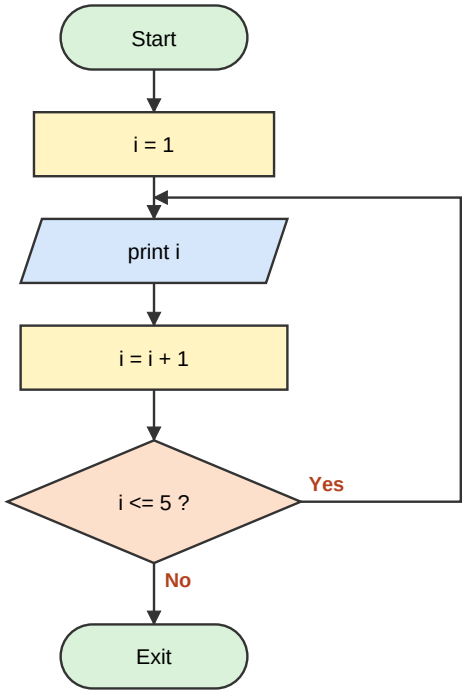
```
int i = 1;
do {
    cout << i << " ";
    i++;
} while (i <= 5);        // output: 1 2 3 4 5
```

Farq: i = 10 ho aur condition `i <= 5` ho to **while** ek baar bhi nahi chalega, par **do-while** ek baar chalega (10 print hoga).

Q9: 1 se 5 print (do-while)

- Pehle print, phir i badhao, phir condition.
- Yes par loop wapas upar.

```
i = 1
do
    print i
    i = i + 1
while i <= 5
exit
```



9. Nested Loops

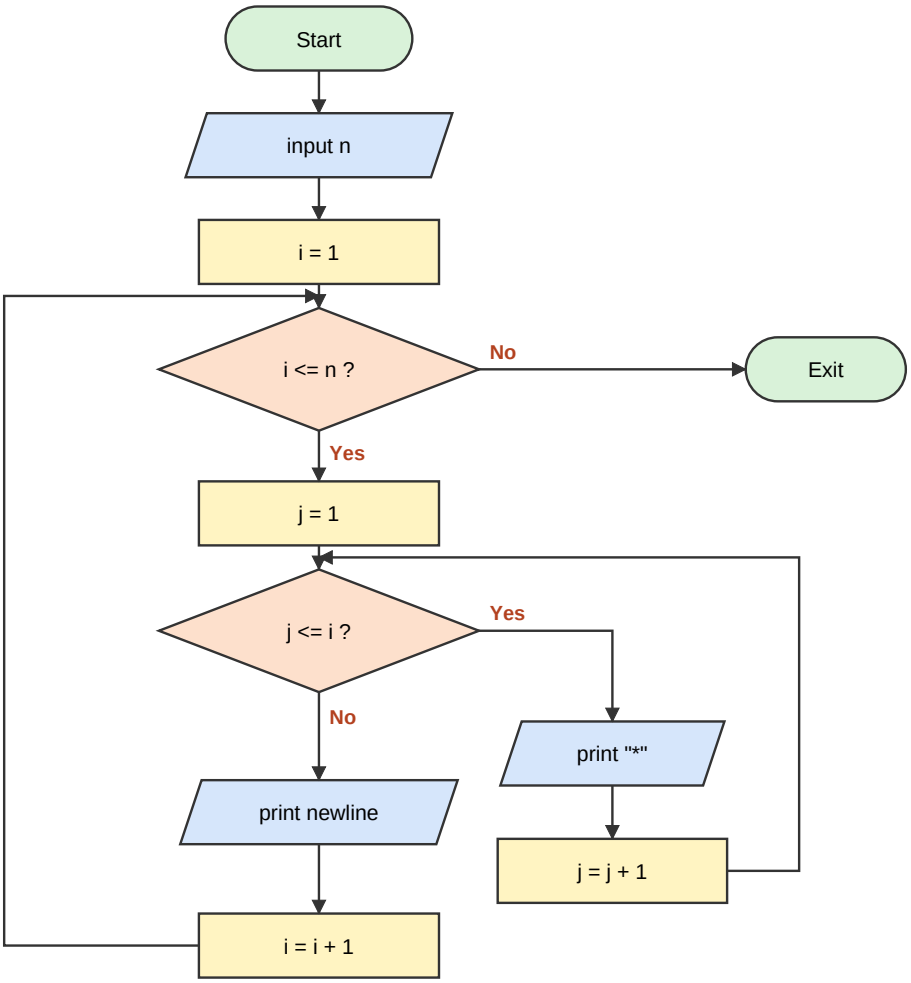
- Loop ke **andar loop**. Outer loop ke **har ek** iteration par inner loop **poora** chalta hai.
- Total iterations = outer x inner (outer 4 baar, inner 3 baar -> 12).
- Pattern printing mein use hota hai: **outer = rows**, **inner = columns**.
- Inner loop ke baad **newline** print karte hain taaki agli row nayi line par aaye.

```
for (int i = 1; i <= 4; i++) {       // rows
    for (int j = 1; j <= i; j++) {   // columns
        cout << "*";
    }
    cout << endl;
}
// *
// **
// ***
// ****
```

Q10: Star triangle (n rows)

- Row i mein **i stars**: isliye inner condition **j <= i**.
- j har row ke liye **dobara 1** se shuru hota hai.

```
input n
for i = 1 to n
    for j = 1 to i
        print "*"
    print newline
exit
```



10. while vs for vs do-while

Loop	Condition kab check	Kab use karein
while	Pehle (entry se)	Iterations ki ginti pata na ho, condition par chalna ho
for	Pehle (entry se)	Ginti pata ho; init, condition, update ek line mein
do-while	Baad mein (kam se kam 1 baar chalta hai)	Pehle kaam, phir dekho dobara karna hai ya nahi

11. Yaad rakhne wali baatein

- Decision = **if / else if / else**. Pehli true condition ka hi block chalta hai.
- **==** compare, **=** assign. **&&** sab true, **||** koi ek true.
- Ternary: `cond ? a : b`.
- Loop ke 3 hisse: **init, condition, update**. Update bhoole to infinite loop.

- Sum ke liye **sum = 0**, factorial ke liye **fact = 1** se shuru.
- do-while kam se kam **ek baar** chalta hai, aur **;** end mein.
- Nested loop: outer ke har iteration par inner poora chalta hai.

12. Practice (khud karo)

- Voting eligibility (age >= 18).
- Leap year check.
- 1 se n tak **even numbers ka sum**.
- Number ke **digits ginna** (hint: n ko 10 se baar-baar divide karo jab tak 0 na ho).
- Rectangle star pattern (3 rows x 5 columns).